

# The 'Incredible' Kamal Mistry

The man behind the year's most talked about animation movie takes us behind the scenes



Photograph: Jiten Gandhi

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**Y**ou have seen the movie, you fell in love with the characters, now meet the man behind *The Incredibles*—Kamal Mistry, the Gujarat-born, New Zealand-educated technical director at Pixar. Founded by Apple CEO Steve Jobs, Pixar is undoubtedly the world's leading animation studio. Mistry speaks on what goes into the making of an animation blockbuster.

## The Magic Is The Story

"At Pixar, we concentrate on the story. Next comes the storyboard. Each and every frame is first drawn there. Then come story layout, modelling, shading, building characters, building models, sets and props, hair shading, and lighting."

## Making Hair

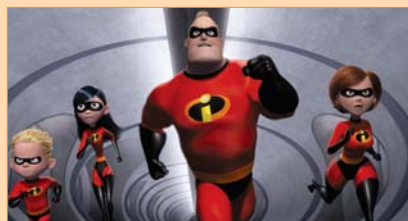
"I was mainly involved in the creation of the hair for the characters. We wanted a 'cartoony' yet real look for the hair. Modelling was done in Maya, including the sets and props. We used Marionette, an in-house animation software, to make hair. We first made key hair strands and then wrote the code that made the hair in between. The next step was to make it

wavy. For this, we made some key strands on the model so we could understand what it would be like. Maya already had tools to comb hair, so that was easy."

## A Hairy Syndrome

"Syndrome (a character in *The Incredibles*) had the most complicated hair—it always stood on end. So I first made a tube first with the key strands, and then filled it with the other hair. This was mainly accomplished with the help of MEL script." (MEL stands for Maya Embedded Language and is used to create scripts that automate processes in Maya). "With MEL, we could create programs that performed tasks within the software. For hair, we wrote many scripts that helped us groom and sculpt hair on all characters."

"The next step was the simulation of the hair we had created. Before Voilet, hair beyond shoulder length was theoretical. Getting the right simulation for her hair was a real challenge. Initially, we tried using cloth strips for hair, but it is



unsuitable as cloth behaves differently. It was then worked out with the help of velocity smoothing algorithm and our own simulator program."

## Giving Characters A Soul

"For the character articulation of *The Incredibles*, we developed software that gave us the ability to create bones and muscles, and then skin that would sit on the bones and muscles! We had the ability to tweak the bones, the muscles as well as the skin. This was necessary to create the look of characters like 'Mr Incredible'.

"Character articulation was a real challenge. For 'Edna', we had to do a variety of facial expressions. To get the required outcome, we made a bone-muscle-skin set up where we preserved the volume of muscle as they contracted and expanded. In the muscle-bone system, the muscle would drive the skeleton and the end-result drives the skin. In order to avoid mathematical complications, we made characters that could be manipulated in real-time. Our emphasis for the facial expressions were on clean lines. Director Brad Bird told us to push the character to do as many things as possible."

## Bam! Boom!! Kapow!!!

"After working on hair and characters, the next task assigned to me was the creation of explosions. The director as well as the Art Department wanted to have explosions with 'tentacles' or 'fingers'—where the explosion would start from one point and explode outwards with the tentacle or fingery look. We developed an explosion simulation set up within Maya, where we would draw curves to denote the direction and then simulate the explosion, given those curves. This was all possible with the help of MEL scripts within Maya. "We mostly used Maya's fluid dynamics for explosions. For this, colour heat value and density was considered. Explosions were done using particle dynamics and fluid simulation."

## In A Nutshell

"Making *The Incredibles* was a lot of hard work and fun. Today, when we look back, there is a sense of achievement. My favourite scene in the movie is when Dash gets to run really fast and in the process moves through many sets like forest, land. There was a lot complexity involved, the motion blur, the blur on his feet; it was all very daunting."

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extremely supportive of my work and I've been able to focus on my career." And what has ten years in the animation industry taught her? "I think women have more patience than men!"

## The Final Frame

Synergy would best describe the work environment at a place like Crest. Like dedicated worker ants, heads buried in their workstations, the 250-member strong Team Crest works together to create the "picture perfect scene". Ideas are generated on the go, thrashed at

meetings, and only the best ones make a successful journey to the big screen. "It takes us about a month to create a single, 22-minute episode!" says Dhaliwal. His impish smile does not give away the stress, the late nights, the intense pressure of deadlines... yet, at the end of the day, nothing beats the satisfaction of a job well done.

As we walk away from Crest, Sherwin, our photographer, remarks thoughtfully, "At 24, these guys are in a prime spot." You bet. ■

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